

ConvNet Size & Parameters

Advanced · STUDENT - fill in the blanks

A color image through a deeper ConvNet - no number repeats, so lean on the formulas.

THE ONLY TWO FORMULAS (stride 1, no padding)

Conv2D: $\text{params} = ((M \times N \times L) + B) \times F$ $\text{new size} = \text{old size} - (M - 1)$
 MaxPool: $\text{params} = 0$ $\text{new size} = \text{old size} / \text{pool}$
 Flatten: $\text{params} = 0$ $\text{size} = \text{rows} \times \text{cols} \times \text{channels}$
 Dense: $\text{params} = (\text{inputs} + 1) \times \text{nodes}$ $\text{size} = \text{nodes}$

M, N = kernel size L = channels coming IN (inherited, never chosen)
 B = bias = 1 F = filters = feature maps coming OUT (you choose)

THE CODE (given)

```
model = models.Sequential()
model.add(layers.Conv2D(16, (5, 5), activation='relu',
                        input_shape=(64, 64, 3)))
model.add(layers.MaxPooling2D((2, 2)))
model.add(layers.Conv2D(32, (3, 3), activation='relu'))
model.add(layers.MaxPooling2D((2, 2)))
model.add(layers.Conv2D(64, (3, 3), activation='relu'))
model.add(layers.MaxPooling2D((2, 2)))
model.add(layers.Flatten())
model.add(layers.Dense(128, activation='relu'))
model.add(layers.Dense(1, activation='sigmoid'))
```

Layer	Output shape	Params	Work it out - fill the blanks
Conv2D: F=16, (5x5)			size = $64 - (5-1) = \underline{\hspace{1cm}}$ params = $((5 \times 5 \times \underline{\hspace{1cm}}) + 1) \times 16 = \underline{\hspace{1cm}}$ L=? RGB
MaxPool 2x2			size = $\underline{\hspace{1cm}} / 2 = \underline{\hspace{1cm}}$ params = 0
Conv2D: F=32, (3x3)			size = $\underline{\hspace{1cm}} - (3-1) = \underline{\hspace{1cm}}$ params = $((3 \times 3 \times \underline{\hspace{1cm}}) + 1) \times 32 = \underline{\hspace{1cm}}$ L=?
MaxPool 2x2			size = $\underline{\hspace{1cm}} / 2 = \underline{\hspace{1cm}}$ params = 0
Conv2D: F=64, (3x3)			size = $\underline{\hspace{1cm}} - (3-1) = \underline{\hspace{1cm}}$ params = $((3 \times 3 \times \underline{\hspace{1cm}}) + 1) \times 64 = \underline{\hspace{1cm}}$ L=?
MaxPool 2x2			size = $\underline{\hspace{1cm}} / 2 = \underline{\hspace{1cm}}$ params = 0
Flatten			size = $\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$ params = 0
Dense(128)			params = $(\underline{\hspace{1cm}} + 1) \times 128 = \underline{\hspace{1cm}}$ size = 128
Dense(1, sigmoid)			params = $(\underline{\hspace{1cm}} + 1) \times 1 = \underline{\hspace{1cm}}$ size = 1
TOTAL trainable parameters			add up the Params column

Each row shows its formula with blanks - fill L (inherited) and the results, then check against model.summary().